MIL-G-11491E
19 February 1988
SUPERSEDING
MIL-G-11491D
5 May 1977

MILITARY SPECIFICATION

GLASSWASHER, ELECTRIC, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of portable glasswasher with electric motor driven rotating brushes (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7320

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SPECIFICATIONS

FEDERAL

CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
QQ-A-601 - Aluminum Alloy Sand Castings
PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-461 - Electromagnetic Interference Charactersitics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of

* (Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of solicitation.

UNDERWRITERS LABORATORIES INC. (UL)

UL 921 - Commercial Electric Diswashers

(Application for copies should be addressed to the Underwriters Laboratories, Inc. 333 Pfingsten Road, Northbrook, IL 60062.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip
A 493 - Stainless and Heat-Resisting Steel for Cold Heading and Cold Forging Wire
* D 3951 - Standard Practice for Commercial Packaging

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

WD-1-General Purpose Wiring Devices

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Standard product. The glasswasher shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modification to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturers or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

* 3.3 Codes and Standards. The glasswasher shall comply with the applicable requirements of *Underwriter's Laboratories Standard No. 921 (See 4.4.6 and* 4.4.7).

* 3.4 Data name plate. The glasswasher shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data name plate shall be made of minimum 20 gauge corrosion resisting metal, and attached to glasswasher by rivets or screws in such a manner as to meet the applicable National Sanitation Foundation requirements for this equipment. The plate shall bear the following information which shall be stamped, engraved, or applied by photosensitive means:

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National stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (If different from manufacturer.)
Manufacturer's Model Number
Government Approved Manual Number (see 6.2)

Each plate shall be placed so as to be readily visible to the operator during normal operation and use, and so as to not adversely affect the life or utility of the glasswasher.

3.5 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial glasswasher provided the completed item complies with all provisions of this specification (See 6.4).

3.5.1 Stainless steel. Stainless steel sheet shall conform to any type in the 300 or 400 series of ASTM A 167 or ASTM A 176, respectively. Stainless steel wire shall conform to any type in the 300 series of ASTM A 493.

3.5.2 Aluminum castings. Aluminum castings shall be alloy B 443.0, temper F, or alloy 208.0, temper F or T55 conforming to QQ-A-601.

3.6 Design and construction. The glasswashers shall be designed to remove foreign matter on the inside and outside of two items of glassware simultaneously and at a rate of at least 2000 glasses per hour (see 4.5.1). The unit shall be designed so it can be suspended from a sink splashback above water or used on a drainboard. The glasswasher shall primarily consist of an electric motor, brush drive mechanism, interchangeable brushes, and a splashguard. The glasswasher shall be not more than 20 inches wide by 20 inches high by 18 inches deep.

3.6.1 Drive. Brushes shall be horizontally driven by chain, V, double V, flat or round belts, and pulleys, or gears and pulleys. Belts shall be adjustable and gears shall have machined teeth. The mechanism shall rotate brushes at a minimum speed of 650 revolutions per minute (see 4.5.1). Permanently sealed ball bearings of self-lubricating bronze bearings shall be used for brush drivers.

3.6.1.1 Housing. The drive mechanism shall be enclosed in a housing constructed of material specified in 3.5.1 or 3.5.2 and shall have a means of access to the drive mechanism for adjustment and replacement.

3.6.2 Splash guard. The splash guard shall be constructed of not less than 0.0335-inch thick stainless steel specified in 3.5.1. The guard shall be located above the brushes and shall be formed to deflect water thrown from the

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brushes down into the sink or drainboard. Minimum clearance between the outside diameter of the brushes and the guard shall be 2 inches. All edges shall be finished smooth.

3.6.3 Bracket. The bracket for suspending the glasswasher from the sink splashback shall be fabricated from not less than 0.093-inch thick stainless steel specified in 3.5.1.

* 3.6.4 Brushes. Bristles shall be nylon not less than 0.006 inch diameter for extension brushes and not less than 0.014 inch diameter for center and side brushes. Bristles shall be firmly assembled to the core of the brush to reduce or minimize loss of bristles. Brushes shall be replaceable on the drive shaft and shall have a means for locking in place. Any tools required for brush replacement shall be supplied by the manufacturer. Brushes in the quantity and sizes shown in table I shall be furnished with each glasswasher:

* TABLE I. Required brushes

Quantity	Size (inches) Length	Diameter	Brush	Use
1	6 to 8	3-3/4 to 5	Center	
2	5-1/2 to 8	3	Side	Glasses, cups, mugs
1	2	1-3/4	Extension	Salt shakers

3.7 Electrical characteristics. Unless otherwise specified (see 6.2), the glasswasher shall be designed for operation on a 120-volt, single phase, 60 hertz system.

3.7.1 Motor. The motor shall conform to CC-M-1807 and shall be not less than 1/4-horsepower (187 W), splashproof or waterproof, continuous duty, with thermal overload protection.

3.7.2 Switch. Motor switch shall be an "on-off" type, either single or double pole located on the motor or drive-mechanism housing.

* 3.7.3 Cord and plug. A minimum 6-foot, UL type SJ, SJO, SJT, or SJTO, three-wire flexible cord with grounding conductor and a NEMA 5-15P plug shall be provided. The cord shall be permanently connected to the glasswasher.

3.7.4 Electromagnetic compatibility. When specified (see 6.2), the glasswasher shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461.

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3.8 Performance. The glasswasher shall operate as intended without vibration or noticeable slowing of the brushes and shall remove all visible foreign matter when tested as specified in 4.5.1.

3.9 Finish. The glasswasher shall be finished in accordance with the manufacturer's standard practice.

3.10 Marking.

* 3.10.1 Identification. Marking shall be applied in accordance with MIL-STD-130 (see 3.4).

3.10.2 Instructions. The glasswasher shall be permanently marked on the unit or on a securely attached corrosion-resistant metal plate with the words "Intended for removal of foreign matter. Use dishwasher for adequate sanitation."

* 3.11 Workmanship. Components shall be free from burrs and rough edges. Welds shall be sound, smooth, and free from fractures or fissures. Spot welds shall be free from pits, burn-throughs, and flashes. Scale and flux shall be removed. Washers or lockwashers shall be provided in accordance with standard commercial practice and all bolts, nuts, and screws shall be drawn tight. The glasswasher shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

* 4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3, and tested for the characteristics specified in 4.4.4. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.1.2 Certification. The contractor shall furnish a certificate of compliance for the requirement specified in 3.7.4.

* 4.4.2 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of glasswashers. The sample unit shall be one glasswasher. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturer's standard product except for changes necessary to comply with specification's requirements	X	
Construction and workmanship (applicable to all components)	Any burr on rough edge	X	
	Any component fractured or malformed	X	
	Any component not in proper alignment or not located as specified	X	
	Functioning component that is inoperative or will not operate as intended	X	
Welding	Missing, incomplete, or cracked	X	
	Scale and flux deposits not removed		X
	Weld not smooth or contains pits or burn-through	X	
Threaded fasteners	Missing, broken, or stripped	X	
	Washers and lockwashers not provided where required		X
Detailed construction cord and plug	Missing or not permanently attached	X	
	Not 3-wire grounding type	X	
Housing	Housing not sealed	X	
	Access not provided for drive mechanism	X	
Splash guard	Not formed as specified	X	
Brushes	Not wrapped as specified	X	
	Center brush core not drilled to receive extension brush	X	
	Extension brush does not fit hole or is excessively loose	X	
	Bearings not type specified	X	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Identification marking	Missing, incomplete, not legible		X
Instructions	Not located as specified		X
	Missing, incomplete, not legible	X	
* Data name plate	Omitted or not as specified		X
	Information incomplete or illegible		X
	Not located so as to be readily visible to the operator		X

* 4.4.3 End item dimensional examination. The end items shall be examined for conformance with dimensions specified in 3.6 and table I. The lot size shall be expressed in units of glasswashers. The sample unit shall be one glasswasher with all supplied brushes. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

* 4.4.4 End item testing. One glasswasher from the lot shall be tested as specified in 4.5.1, and when applicable 4.5.2. Tests shall be conducted using electrical characteristics as specified in the contract. Failure to pass any test shall be cause for rejection of the lot.

* 4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling. Bulged or distorted container.

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Examine	Defect (cont'd)
Content	Number per container is more or less than required.

* 4.4.6 Certification compliance examination. Certifications, certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.3., shall be examined and validated as proof of compliance.

* 4.4.7 Underwriter's laboratories, inc. (UL). Acceptable evidence of meeting the applicable requirements of the Underwriter's Laboratories standard specified in 3.3 shall be the UL label, a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the glasswasher conforms to the applicable requirements of the specified UL standards (see 3.3).

* 4.5 Methods of inspection.

4.5.1 Performance test. At least 10 glasses or mugs shall be soiled with cocoa or milk on inside and outside surfaces and allowed to dry. Two mugs or glasses shall be momentarily dipped into warm water and then placed in cleaning position over the two side brushes. The procedure shall be repeated for the remaining items and the cleaning rate, brush speed, and machine operation shall be checked to determine conformance with 3.6, 3.6.1, and 3.8, respectively. Any nonconformance shall constitute failure of this test.

4.5.2 Electromagnetic compatability test. When electromagnetic compatability is required, the glasswasher shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD 462. The Government reserves the right to witness tests performed by the contractor or his private testing organization. The contractor shall furnish to the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Compatibility Test Report and requirements meet MIL-STD-461. Any nonconformance shall constitute failure of this test.

5. PACKAGING

* 5.1 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.1.1 Level A packing. Each glasswasher shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Fiberboard inserts shall be used to immobilize the contents. Uninstalled brushes shall be secured in an unused portion of the shipping container in such a manner as to prevent movement or damage while in transit. Each shipping container shall be closed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

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* 5.1.2 Level B packing. Each glasswasher shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or type SF, class domestic, grade 275 of PPP-B-636. Fiberboard inserts shall be used to immobilize the contents. Uninstalled brushes shall be secured in an unused portion of the shipping container in such a manner as to prevent movement or damage while in transit. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.1.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

* 5.1.3 Commercial. Glasswashers shall be packed in accordance with ASTM D 3951.

5.2 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The glasswasher is intended for removing soiled matter from glassware prior to washing and sanitizing in a dishwashing machine. The glasswasher is not intended for immersion in a sink.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, 6.3).
* c. Government approved manual number to be included on Data Name Plate (see 3.4).
d. Electrical characteristics, if other than specified (see 3.7).
e. When electromagnetic compatability is required (see 3.7.4).
f. Selection of the applicable level of packing (see 5.1).
g. When weather-resistant class fiberboard shipping containers are required for level B packing (see 5.1.2.1).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.5).

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* 6.5 Subject (key word) listing

Glasswasher, electric
Washer

* 6.6 Changes from previous issue. The margins of this document are marked with an asterisk(*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. this was done as a convenience only and the Government assumes no laiability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 99

Review Activity:

Army - MD
* Air Force - 84

User activities:

Navy - CG, SA

Preparing activity:

Army - GL

Project No. 7320-0821

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-11491E	GLASSWASHER, ELECTRIC, PORTABLE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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